THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL
FORM FOUR PRE-MOCK EXAMINATION
033 BIOLOGY

Time: 3 Hours **April- 2025**

Instructions

1. This paper consists of sections A, B and C.
2. Answer ALL questions from section A and B and only two questions from section C.
3. All writings must be written in blue or black ink.
4. Cellular phones, calculators and other unauthorized materials are not allowed in the examination.
5. Write your Examination Number on every page of your answer sheets

SECTION A (16 MARKS)

Answer **all** questions in this section

1. For each of the items (i)-(x), choose the most correct answer among the given alternatives and write its letter beside the item number.

   i. In which of the following structures is the Bowman's capsule found?
   A. Neurone
   B. Myosin
   C. Xylem
   D. Nephrone
   E. Phloem

   ii. When doing an experiment to determine the dry weight of a food sample, Seypalm heated the food sample and then weighed it repeatedly until when he obtained a constant weight. In which apparatus should he store the dried food sample in order for its weight to remain the same?
   A. Desiccator
   B. Incubator
   C. Petri dish
   D. Crucible
   E. Pipette

   iii. One of the following is the effect of high cholesterol, particularly LDL cholesterol:
   A. Can lead to build up of sugar in the blood, and increasing the risk of diabetes
   B. Can lead to build up of sugar in the blood, and increasing the risk of tuberculosis
   C. Can lead to build up of salts in the blood, and increasing the risk of kidney stones
   D. Can lead to build up of plaque in arteries and increasing the risk of heart disease
   E. Can lead to build up of proteins in the blood, and increasing the risk of Kwashiorkor

iv. The ability of organisms to maintain a relatively constant or stable internal environment is correctly defined as:
A. Entomology
B. Excretion
C. Endocytosis
D. Regulation
E. Homeostasis

v. Seypalm's child was playing with a razor blade when she accidently cut herself. She suffered just a minor injury but still some red liquid flowed out of the wound before drying up. The substance that flowed out of her wound is most likely to be:
A. A system
B. A tissue
C. A cell
D. A mucus
E. A living thing

vi. Some earlier scientists believed that from non-living matter, living matter developed. For example; worms and frogs could have come from mud or rotten food. This is……..
A. Cosmozoic theory
B. Theory of spontaneous generation
C. Theory of organic evolution
D. Steady state theory
E. Theory of the survival of the fittest

vii. Immediately after fertilization has taken place inside the ovary of the flower of a plant, the fertilized ovule is known as:
A. Embryo
B. Radicle
C. Zygote
D. Plumule
E. Fruit

viii. One of the followings sets contains hormones that are produced in the gonad only, which is it?
A. Testosterone, Prolactin and Thyroxine
B. Anti-diuretic hormone, Progesterone and Thyroxine
C. Prolactin, Insulin and Glucagon
D. Oestrogen, Testosterone and Progesterone
E. Insulin, Oestrogen and oxytocin

ix. Which of the following statement is true about amoeba and paramecium?
A. Some amoebae are aquatic organisms while all paramecia are terrestrial organisms
B. Paramecia are macroscopic organisms while amoebae are microscopic organisms
C. Both amoebae and paramecia are macroscopic organisms
D. Most paramecia are larger than amoebae and can feed on amoebae
E. Most amoebae are larger than paramecia and can feed on paramecia

x. Which of the following are nitrogenous bases in DNA arranged in their correct pairing?
   A. Adenine with Uracil, Guanine with cytosine
   B. Guanine with Thymine, Uracil with cytosine
   C. Adenine with Thymine, Guanine with cytosine
   D. Adenine with Guanine, Thymine with cytosine
   E. Adenine with cytosine, Guanine with Uracil

2. Match the uses of the components of the First Aid Kit in LIST A with the corresponding components in LIST B by writing the letter of the correct response beside the item number in the answer booklet provided.

LIST A	LIST B
i. They are used for cutting bandages and pieces of clothes. ii. It is used to soften a minor injury. iii. Are tied with bandages to support broken bones. iv. For holding bandages together. v. For cleaning wound by killing germs. vi. For removing thin objects that pierced the skin.	A. Methylated spirit B. Safety pins C. Rolls of adhesive tape D. Tweezers E. A pair of scissors: F. Splints G. Petroleum jelly H. Cotton wool I. Liniment J. Elastic bandages

SECTION B (54 MARKS)

Answer all questions in this section

3. Form four students at Longido Secondary School were arguing whether they themselves were examples of organisms under the phylum Chordata. With one example in each case, briefly explain to them six classes of the phylum Chordata so that they can know whether they are also animals belonging to the phylum Chordata.

4. It is important to learn about waste so that to know the best way of managing it. Educate the public about classification of wastes according to the physical states and explain the basic principles of waste management.

5. Blood performs many important functions within the body; explain any six of such functions.

6. Outline six factors that affect growth in plants.

7. (a) Plants do not have complex excretory organs like those of animals. Give three reasons to explain this observation.

(b) The following table displays excretory products, metabolism, excretory structures and example of organisms associated with the excretory products. Fill any missing information to complete the table.

Excretory Products	Metabolism	Excretory structure	Example of Organism
(a) Carbon dioxide and water			Man, frog, bird
		Stomata	Plants
	Photosynthesis		Plants
Ammonia	Protein metabolism		Amoeba
Urea			Man
(b) Uric acid			Lizard, bird
(c)	Hemoglobin breakdown		Man

8. Dominance is a state of one character or gene from one parent masking the corresponding character from another parent. Briefly describe the three types of dominance.

(b) A red flowered rose plant was crossed with a white flowered rose plant and all members of F1 were pink flowered plants. When the pink flowered rose plants were self-crossed, a mixture of red, pink and white flowered plants were obtained.

i. Make a cross to show the genotype of F1.
ii. Make a cross to show the genotype of F2.
iii. State the type of dominance exhibited in the above situation

SECTION C (30 MARKS)

Answer two (2) questions from this section.

9. Because of their essential roles in the digestion of food material, our mouths need special care and attention in order to avoid dental caries. As a form four student who studied Biology very well, educate the public on how the human mouth is adapted for digestion of food and the causes, symptoms, and control of dental caries.
10. Many countries have declared war against drug abuse and addiction, and those countries work together in trying to eradicate the problem. Explain at least six effects of drug abuse and addiction that led to so many countries considering it as a very bad problem, and suggest at least four measures to control the drug abuse.
11. People suffering from HIV/AIDS need special care and support. In four points, give the reasons of providing care and support to PLWHA?